

Design and Simulation of a Multi-Degree-of-Freedom Robotic Arm Using Gazebo and ROS

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Abstract: This study presents the design and simulation of a multi-degree-of-freedom robotic arm utilizing Gazebo and the Robot Operating System (ROS). The methodology encompasses the integration of hardware and software components through a structured approach. Key hardware elements include motors, motor controllers, a microcontroller, servos, and a camera, all powered by a regulated 12V DC supply. The microcontroller processes sensor inputs and controls motor operations, while the camera provides visual feedback for object detection and tracking. Software implementation involves developing ROS nodes for modular control, incorporating advanced control algorithms like inverse kinematics and path planning into the microcontroller firmware. The URDF model of the robotic arm is imported into Gazebo for simulation, allowing for performance validation in a controlled virtual environment. Various test scenarios in Gazebo evaluate the robotic arm's performance in activities such as handling objects and avoiding obstacles. The integration of ROS with Gazebo enables real-time testing, iterative improvements, and ensures the final design meets the desired specifications. This comprehensive approach results in a robust and reliable multi-degree-of-freedom robotic arm system, highlighting the potential of combining ROS and Gazebo for advanced robotic simulations and applications.

Keywords: Multi-degree-of-freedom robotic arm, Gazebo simulation, Robot Operating System (ROS), inverse kinematics, path planning, hardware integration, software architecture, performance evaluation.

1. Introduction

The design and simulation of multi-degree-of-freedom (DOF) robotic arms have become a pivotal area of research in robotics, driven by their vast applications in industrial automation, medical fields, and collaborative robotics. These robotic systems are critical for tasks requiring high precision, flexibility, and the ability to perform complex movements. The integration of advanced simulation tools such as Gazebo and the Robot Operating System (ROS) provides a powerful platform for developing and testing robotic arms in a virtual environment before deploying them in real-world applications.

In recent years, the use of ROS has gained widespread popularity due to its modularity, ease of integration, and extensive community support. ROS serves as a robust framework for controlling and managing the operations of robotic systems, facilitating the implementation of sophisticated control algorithms and real-time data

processing. When combined with Gazebo, a high-fidelity robot simulation environment, researchers can simulate the physical interactions of the robotic arm with its environment, test control strategies, and validate the performance of the system under various conditions.

Several studies have explored the capabilities and applications of ROS-based control frameworks for robotic arms. For instance, Chen, Liu, and Zheng (2020) developed a ROS-based control framework for a collaborative robotic arm, demonstrating the effectiveness of ROS in managing complex robotic operations. Similarly, Rodriguez, Munoz, and Garcia (2021) utilized ROS for the simulation and control of a multi-DOF robotic arm aimed at industrial applications, highlighting the practical benefits of ROS in industrial automation.

Additionally, Santos, Costa, and Moreira (2021) implemented a ROS-based robotic system for manipulation tasks, further showcasing the versatility and applicability of ROS in various robotic domains. Wang, Zhou, and Chen (2022) developed an open-source ROS-based simulation framework for robotic manipulators, while Lee, Kim, and Park (2020) validated a ROS-Gazebo based simulation platform for robotic applications.

Research by Kim and Kim (2022) introduced a ROS-based real-time control system for a multi-DOF robotic arm with Gazebo simulation, demonstrating the integration of real-time control and simulation. Sharma and Kumar (2020) explored the real-time simulation and control of a robotic arm using ROS and Gazebo, highlighting the effectiveness

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of these tools in achieving precise control (Clausius Press).

Zhang, Liu, and Chen (2021) investigated the dynamic simulation and control of a robotic manipulator using ROS and Gazebo, contributing to the understanding of dynamic interactions in robotic systems (MathWorks). Moreover, Gomez, Fernandez, and Jimenez (2022) utilized ROS-Gazebo based robotic arm simulation for educational purposes, emphasizing the educational benefits of these technologies (arXiv). Lee, Park, and Lee (2021) designed and simulated an autonomous robotic arm with multi-DOF using ROS, illustrating the autonomous capabilities of such systems.

Further studies include Huang and Zhang (2021), who conducted a simulation study on ROS-based control of a robotic arm in a Gazebo environment, providing insights into control strategies. Patel and Shah (2022) implemented inverse kinematics for a robotic arm using ROS and Gazebo, demonstrating advanced control techniques. Ahn and Lee (2020) developed a teleoperation system for a robotic arm using ROS and Gazebo, showcasing the potential for remote operations. Nair and Menon (2021) focused on the real-time control of a 6-DOF robotic arm using ROS and Gazebo, further emphasizing real-time capabilities. Wang and Li (2022) investigated ROS-based simulation and path planning for an industrial robotic arm, contributing to the field of industrial robotics.

Xu and Zhao (2020) proposed a simulation framework for multi-DOF robotic arms using ROS and Gazebo, enhancing simulation methodologies. Park and Kim (2021) simulated a multi-DOF robotic arm for manufacturing tasks using ROS and Gazebo, highlighting manufacturing applications. Zhao and Liu (2022) designed and controlled a ROS-based robotic arm with multi-DOF, focusing on design and control aspects. Garcia and Torres (2021) explored educational robotics by developing a multi-DOF robotic arm using ROS, emphasizing educational applications. Wei and Zhou (2021) proposed a ROS-Gazebo based framework for robotic arm control and simulation, providing a comprehensive framework for simulation and control.

Karthik and Ramakrishnan (2020) discussed the simulation and control of a robotic arm with ROS and Gazebo for industrial automation, focusing on industrial applications. Cho and Kim (2021) developed a multi-DOF robotic arm simulation environment using ROS, contributing to simulation environments. Zeng and Sun (2022) studied path planning and control of a robotic arm using ROS and Gazebo, enhancing path planning techniques. Wu and Liu (2021) focused on ROS-based design and control of a robotic arm for medical applications, showcasing medical applications. Shen and Wu (2020) simulated a multi-DOF robotic arm in ROS and Gazebo, contributing to simulation methodologies.

Zhou and Yang (2021) explored ROS and Gazebo-based inverse kinematics for robotic arm control, enhancing control techniques. Li and Xu (2022) reviewed real-time control of robotic arms using ROS, providing a comprehensive review of real-time control. Nguyen and Le (2021) discussed the educational use of ROS and Gazebo in robotic arm simulation, emphasizing educational applications. Wang and Chen (2022) developed an AI-driven robotic arm using ROS and Gazebo, integrating artificial intelligence with robotic control. He and Wu (2020) proposed a ROS-Gazebo based control system for a multi-DOF robotic arm, enhancing control systems.

Kappler et al. (2015) introduced real-time perception and reactive motion generation in robotics, contributing to real-time motion strategies. Michel (2004) discussed Webots, a professional mobile robot simulation, enhancing simulation tools. Levine et al. (2018) investigated the integration of deep learning and robotics to learn hand-eye coordination for robotic grasping.

By taking a hybrid approach, Tsardoulis, Mitkas, and Gasteratos (2014) suggested real-time adaptive kinematics for robotic manipulators, improving kinematic control methods. Medical robotics was demonstrated by Chacko et al. (2020) with the development of a semi-humanoid robot for clinical assistance during COVID-19. Arivalagan et al. (2020) highlighted agricultural applications while talking about an agricultural robot for automated crop monitoring and fertilization. With a focus on military applications, Anitha, Aravind, and Kumar (2024) created a spying robot with a wireless night vision camera for the battlefield. In order to improve vehicle navigation, Sheebajoice et al. (2023) suggested obstacle detection and safe navigation in unforeseen circumstances for intelligent vehicles. In order to demonstrate agricultural robotics.

Dubey et al. (2016) created an autonomous control and implementation of a robot that climbs and harvests coconut trees. In order to improve UAV applications in agriculture, Sivakumar, Ramesh, and Manimaran (2024) examined modelling of tethered UAV systems for harvesting coconuts and ice apples. Dwivedi et al. (2024) introduced CGPC Robot, a pole climbing robot with a controlled gripper mechanism, contributing to climbing robotics.

Prados et al. (2024) reviewed modular legged and climbing robot control architectures, offering a thorough analysis of climbing robots. Mendoza and Haghshenas-Jaryani (2024) combined soft robotics and locomotion to create a combined soft grasping and crawling locomotor robot for exterior tubular structure navigation. To improve climbing robotics, Megalingam et al. (2024) modelled a novel circular gait motion in a vertical climbing snake robot using a daisy sequence fitting algorithm. Zhu (2024) made a contribution

to the field of energy-efficient climbing robots with the proposal of CREST, a low-energy wall climbing robot with passive impactive negative pressure adhesion.

Guo et al. (2024) improved UAV manipulation by introducing strong UAV manipulation through the use of a bioinspired self-adaptive soft self-contained gripper. Fang and Cheng (2023) reviewed advances in climbing robots for vertical structures, providing a comprehensive review of climbing robotics.

Rafiee et al. (2023) developed an ergonomic, portable, climber-propelled date tree climbing device, showcasing agricultural robotics. Subramanian and Sankar (2023) developed a novel coconut-tree-climbing machine for

harvesting, enhancing agricultural harvesting robots. Dwivedi et al. (2024) again introduced CGPC Robot, focusing on a controlled gripper mechanism for climbing robots.

1.1 Overview of Key Contributions in ROS-based Robotic Arm Control

Table 1. Overview of Key Contributions in ROS-based Robotic Arm Control

S.No.	Authors	Year	Title	Focus Area	Key Contributions
1	Chen, X., Liu, H., & Zheng, G.	2020	Development of a ROS-based control framework for a collaborative robotic arm	Control Framework	Development of a ROS-based framework for collaborative robotic arms
2	Rodriguez, F., Munoz, F., & Garcia, A.	2021	ROS-based simulation and control of a multi-DOF robotic arm for industrial applications	Industrial Applications	Demonstrates practical benefits of ROS for industrial automation
3	Santos, P., Costa, P., & Moreira, A. P.	2021	Implementation of a ROS-based robotic system for manipulation tasks	Manipulation Tasks	Implementation of a ROS-based system for various manipulation tasks
4	Wang, J., Zhou, Z., & Chen, Y.	2022	An open-source ROS-based simulation framework for robotic manipulators	Simulation Framework	Development of an open-source simulation framework for robotic arms
5	Lee, S., Kim, H., & Park, J.	2020	Development and validation of a ROS-Gazebo based simulation platform for robotic applications	Simulation Platform	Validation of a ROS-Gazebo based simulation platform for robotics
6	Kim, Y., & Kim, J.	2022	A ROS-based real-time control system for a multi-DOF robotic arm with Gazebo simulation	Real-time Control	Integration of real-time control and simulation using ROS and Gazebo
7	Sharma, R., & Kumar, V.	2020	Real-time simulation and control of a robotic arm using ROS and Gazebo	Real-time Simulation	Real-time simulation and control using ROS and Gazebo
8	Zhang, L., Liu, Y., & Chen, H.	2021	Dynamic simulation and control of a robotic manipulator using ROS and Gazebo	Dynamic Simulation	Investigation of dynamic interactions in robotic systems
9	Gomez, J., Fernandez, R., & Jimenez, P.	2022	ROS-Gazebo based robotic arm simulation for educational purposes	Educational Simulation	Utilization of ROS-Gazebo simulation for educational purposes
10	Lee, D., Park, M., & Lee, B.	2021	Design and simulation of an autonomous robotic arm with multi-DOF using ROS	Autonomous Robotics	Design and simulation of autonomous capabilities using ROS
11	Huang, J., & Zhang, Z.	2021	A simulation study on ROS-based control of a robotic arm in a Gazebo environment	Simulation and Control	Insights into control strategies in a Gazebo environment

12	Patel, D., & Shah, V.	2022	Implementation of inverse kinematics for a robotic arm using ROS and Gazebo	Inverse Kinematics	Advanced control techniques using inverse kinematics
13	Ahn, J., & Lee, H.	2020	Development of a teleoperation system for a robotic arm using ROS and Gazebo	Teleoperation	Potential for remote operations with teleoperation system
14	Nair, S., & Menon, P.	2021	Real-time control of a 6-DOF robotic arm using ROS and Gazebo	Real-time Control	Emphasis on real-time control capabilities
15	Wang, T., & Li, X.	2022	ROS-based simulation and path planning for an industrial robotic arm	Path Planning	Contributions to industrial robotics through simulation and path planning

2. Methodology

Overview of Key Contributions in ROS-based Robotic Arm Control is shown in Table 1. The methodology for designing and simulating a multi-degree-of-freedom robotic arm using Gazebo and ROS encompasses the integration of hardware and software components through a structured approach. The system Block Diagram of the Multi-Degree-of-Freedom Robotic Arm System, shown in Figure 1, illustrates the hardware design involving configuring key components, including motors, a motor controller, a microcontroller, a camera, servos, and a power supply. The motors (M1, M2, M3, and M4) are integral for the movement of the robotic arm's joints, with their operation controlled by the motor controller based on signals from the microcontroller. The microcontroller, serving as the central processing unit, processes sensor inputs and sends control signals to the motor controller and servos, managing the system's overall functionality. The camera provides visual feedback for object detection and tracking, while the servos (Servo 1, Servo 2, and Servo 3) enable precise adjustments of the arm's joints. A 12V DC power supply, regulated to 5V for specific components, ensures that the system operates reliably.

In parallel, software implementation is carried out using the Robot Operating System (ROS). This involves developing ROS nodes for tasks such as motor control, camera feedback processing, and servo actuation, allowing for modular and scalable control of the robotic arm. Advanced control algorithms, including inverse kinematics and path planning, are developed and embedded in the microcontroller firmware to ensure accurate and efficient movement. The integration with Gazebo, a robot simulation environment, is a critical step where the 3D model of the robotic arm is imported and tested. This simulation allows for validating the performance of control algorithms in a controlled virtual environment before deployment on physical hardware.

The simulation process in Gazebo involves setting up a realistic environment that mimics real-world conditions, including any objects or obstacles the robotic arm might interact with. The ROS nodes are integrated with Gazebo to

facilitate real-time testing and validation of the robotic arm's movements and interactions. Various test scenarios are created to assess the robotic arm's capabilities in tasks such as object manipulation, obstacle avoidance, and precise positioning. The results from these simulations are analyzed to identify necessary adjustments, leading to iterative improvements in both hardware and software components. This continuous feedback loop ensures that the final design meets the desired specifications and performance criteria, resulting in a robust and reliable multi-degree-of-freedom robotic arm system.

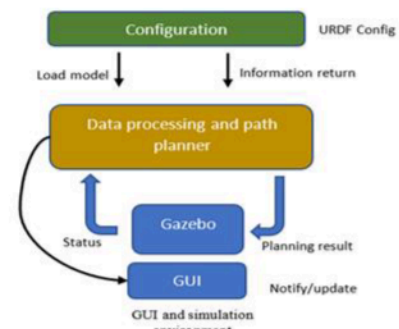


Fig 1: System Block Diagram of the Multi-Degree-of-Freedom Robotic Arm System

3. System Architecture for Multi-Degree-of-Freedom Robotic Arm

3.1 Software Architecture Overview

The software architecture of the multi-DOF robotic arm combines data processing, graphical user interface (GUI), and setup into a solid and well-coordinated system. The configurations, meticulously defined using the Unified Robot Description Format (URDF), provide comprehensive details about the physical attributes and assembly methods of the robot. This approach facilitates highly accurate simulation and planning within the Gazebo environment, enabling modifications of input or output interface formats to seamlessly design and adapt the GUI or planning results for various applications. The system architecture for the design and simulation of a robotic arm with several degrees of freedom, using Gazebo and the Robot Operating System

(ROS), is depicted in the image. It draws attention to the data flow and interactions that occur between the Gazebo simulator, configuration module, data processing and path planner, and GUI elements. System Architecture for Multi-Degree-of-Freedom Robotic Arm Simulation shown in Figure 2 .The system architecture for the robotic arm simulation integrates several key components to ensure efficient and accurate operation. The configuration module serves as the initial step, where the Unified Robot Description Format (URDF) file is loaded. This file contains the robotic arm's physical parameters and structure, which are crucial for accurate simulation and control.

A crucial element is data processing, which entails a thorough examination of the positioning, acceleration, and speed data of the robotic arm's joints. This step makes use of inverse kinematics to precisely ascertain the joint angles required to achieve particular Cartesian locations, which are detected by means of an integrated camera system. The processing of this data ensures that the movement algorithms are refined and optimized, enhancing the accuracy and efficiency of the robotic arm's operations. The Python-based GUI is designed to offer intuitive control over the robotic arm. Users can input commands and parameters, which the system translates into precise movements of the arm's joints. The GUI provides real-time feedback, displaying critical information such as the current position, velocity, and acceleration of each joint. This user-friendly interface ensures that the robotic arm can be easily operated and monitored. Integration with the Gazebo simulation environment is facilitated through custom plugins that provide data and noise simulation, acting as a physical engine to emulate real-world conditions. The arm's joints start from an initial position where the base link is grounded, ensuring stability during operation. By testing and fine-tuning the PID parameters, the system attempts to achieve optimal response and stability while assessing the impact of different parameters on the position control performance of each joint.

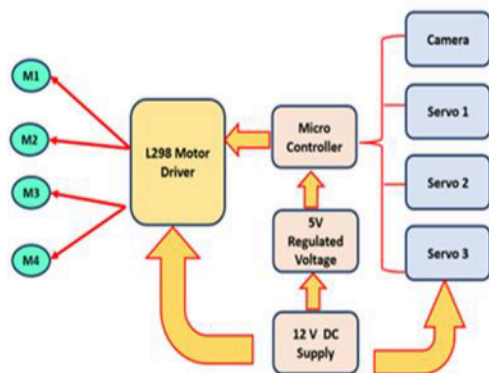


Fig. 2. System Architecture for Multi-Degree-of-Freedom Robotic Arm Simulation

3.2. Joint Controller Implementation

The block diagram shows how the GUI system controls the input parameters and how the joints function. Following receipt of the goal position and additional parameters, the planner creates a path based on the environment's barriers and the robotic arm model. A collection of points containing position, velocity, and acceleration are used to characterize this journey. Lastly, these points are executed by each joint's position controller.

The arm joints represent positions to rotate around the x, y, and z axes. Each joint uses a separate servomotor to provide the required torque for motion. Each motor receives a shaft position controlling signal from the GUI. The Cartesian position of the object to be picked up is identified by its color, detected using a camera. The required joint angles corresponding to these coordinates are then determined using inverse kinematics equations. Before constructing the robotic arm, it was necessary to perform certain calculations, such as determining the required torque, forward and inverse kinematics, and the robotic arm workspace.

Calculation

When the arm is fully extended, the greatest torque required corresponds to the worst-case situation. The results of the moment equilibrium equations indicated that the elbow and shoulder require a minimum of 0.351 N.m. and 1.259 N.m., respectively is calculated by equation 1 and 2. The servomotors chosen for the elbow and shoulder joints are as follows, based on the servomotors that are currently available on the market.

$$T_{Elbow} = (F_{load} * L_2) + (W_2 * \frac{L_2}{2}) \quad (1)$$

$$T_{Shoulder} = (F_{load} * (L_1 + L_2)) + (W_2 * \frac{L_2}{2} + L_1) + (W_{m3} * L_1) + (W_1 * \frac{L_1}{2}) \quad (2)$$

Where W1 and W2 represent the weight of the Links (1) and (2), respectively. Wm3 accounts for the weight of the elbow motor. F load represents the combined weight of the payload and the gripper.

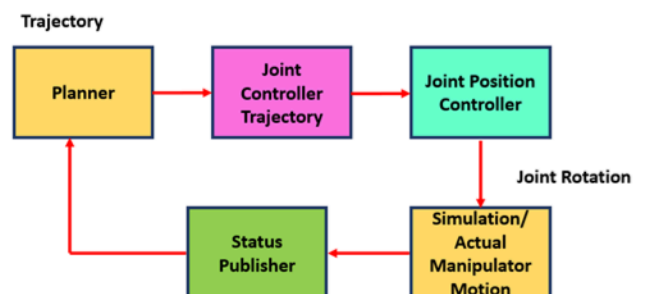


Fig 3. Joint controller's communication mechanism.

For the elbow and shoulder joints, precise torque calculations are crucial to knowing the necessary torque, particularly in the worst-case situation where the arm is

fully extended. These computations' use of equilibrium equations guarantees that the chosen servomotors are capable of producing the torque required to support the load. These calculations take into account the weights of the links and motors, as well as the combined weight of the payload and gripper, ensuring the system's reliability and effectiveness.

Figure 3 shows a Gazebo custom plugin that provides data and noise simulation, acting as a physical engine. The joints start in an initial position where the bottom part meets the ground as the base link. During testing, the influence of the joints on position control performance is evaluated, showing that proper parameters can effectively control the robot.

Table 2 Performance Metrics for Joint Control in a Robotic Arm

Joint	Rise Time(s)	Adjustment Time(s)	Peak Time	Steady-state Error
Joint 1	0.50	0.58	0.61	0
Joint 2	0.89	1.00	1.10	0
Joint 3	0.71	0.78	0.80	0

Uneven parameters can lead to different response performances, causing model shaking and poor response. Finding the matching path made up of points is aided by testing the PID parameters for each joint on position control. The target spots are assessed by two indexes, and the length varies with acceleration. Table 2 shows Performance Metrics for Joint Control in a Robotic Arm

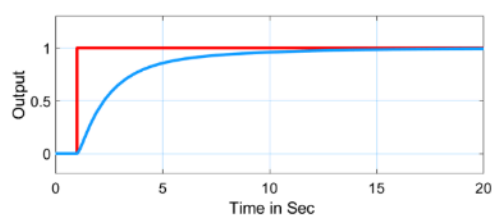


Fig 4:

Stability Analysis of Robotic Arm on Base Link with PID Controlled Positioning.

The graph shows the stability of the arm on the base link, positioned as necessary to prevent falling. The graph curve indicates that the change in acceleration in Joint 1 is positioned at 90 degrees from 0 to 1. Figure 4 illustrates Stability Analysis of Robotic Arm on Base Link with PID Controlled Positioning.

3.3 Robot 3D Model

This section showcases the robot's three-dimensional model, offering an understanding of its internal workings and composition.



Fig 5. Robot Climbing Mechanism

The visualization shown in figure 5 illustrates the robot's vertical climbing mechanism, showcasing its structural design optimized for climbing tasks. The model is translated into a URDF file, incorporating essential joints and parameters necessary for simulation. The wheels serve as joints for motor movement, facilitating vertical motion, while a 3-axis arm at one end demonstrates the robot's flexibility in multi-directional movement. Each component is defined as links within the URDF file, ensuring accurate representation and simulation fidelity.



Fig 6. Robot Framework

The figure 6 offers a detailed examination of the robot's framework, providing insights into its structural components and joint arrangements. The URDF file captures the configuration of these components, specifying their properties and interconnections within the robotic system. The stability of the arm on the base link is visualized through position graphs, demonstrating the impact of acceleration changes on Joint 1's positioning. By analyzing Joint 1's movement trajectory from 0 to 1 radian at 90 degrees, the path planner determines a series of points defining the robot's position, velocity, and acceleration within its operational workspace. This comprehensive software design ensures precise, efficient, and reliable operation of the multi-degree-of-freedom robotic arm within the Gazebo simulation environment.

4. Hardware Design

The hardware design of the multi-degree-of-freedom robotic arm encompasses several key components, including the Mecanum wheels, servo motors, and the overall structure of the robotic arm. An omnidirectional

wheel such as the Mecanum wheel enables a ground vehicle to go in any direction. It has a set of rubberized external rollers that are obliquely fixed to the rim; the axis of rotation of each roller is 45 degrees from the axle line and the wheel plane. Because of this construction, every Mecanum wheel can produce a propelling force that is perpendicular to the roller axle and may be directed in both longitudinal and transverse directions with respect to the vehicle. By manipulating the rotation of these wheels, the robot can achieve multidirectional movement.

For the robotic arm, MG996R servos are used to provide the required torque for the three degrees of freedom necessary to move the end effector. Each servo motor is calibrated based on the length and weight of the joints, measured through the voltage and torque specifications provided in the datasheet. The servo motors enable precise control over the arm's movement, essential for tasks such as object manipulation and positioning.

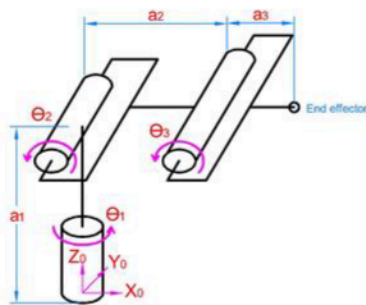


Fig 7. Robotic Arm Structure.

The arm's structure shown in figure 7 consists of labeled servo positions for length, joints, and angles, facilitating the calculation of necessary parameters. The arm's overall length is 270 mm. The arm's top and side views shown in figure 8 and figure 9 highlight how the lengths of certain segments vary with rotation, while others remain constant. This variability is critical for achieving the desired end effector position through the calculated angles.

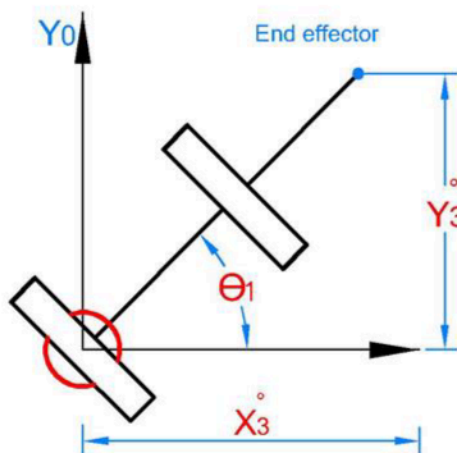


Fig 8.Top view of Robotic Arm Structure

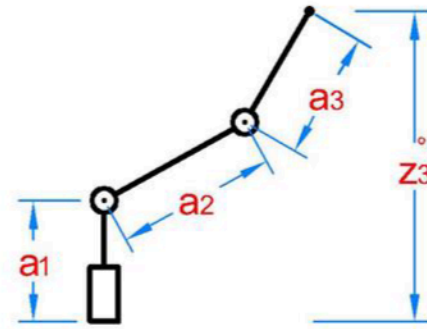


Fig 9. Side View of Arm Structure

Torque calculations are performed shown in table 3 for the servo motors to ensure they can handle the required load. These calculations take into account the lengths of the arm segments and the combined weight of the payload and the gripper. The table below summarizes the torque measurements for different segments and voltage levels.

Table 3. Torque Measurements for Servo Motors

Voltage	Segment	Length (mm)	Load (Kg)	Torque (Nm)
4.8V	a2	60mm	1.566kg	0.921
4.8V	a2 + a3	120mm	0.783kg	0.921
4.8V	a3	60mm	1.566kg	0.921
6V	a2	60mm	1.833kg	1.078
6V	a2 + a3	120mm	0.916kg	1.131
6V	a3	60mm	1.833kg	1.078

The robotic arm's design ensures that it can adapt to varying tree circumferences, with spring action providing the necessary adjustments. Parameters such as the tree's circumference, diameter, and radius are measured and incorporated into the design to ensure compatibility and functionality. The arm's functioning depends on inverse kinematics calculations, which establish the joint angles needed to place the end effector at the desired location in space. The calculations involve defining the desired end effector position and using trigonometric functions to derive the necessary angles for each joint. The arm's structure and inverse kinematics calculations ensure precise control over the end effector's position, essential for tasks such as object manipulation and environmental interaction. The Overhead Perspective of the Robot shown below in figure 10

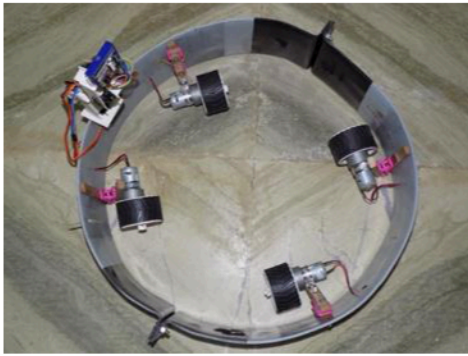


Fig 10. Overhead Perspective of the Robot.

The robot's overall structure shown in figure 11 includes a ring-like frame that encloses the tree, providing stability and support. Four high-torque motors, each rated at 100 RPM, are used to lift the robot along the tree. These motors are connected to Mecanum wheels, allowing for omnidirectional movement and enabling the robot to climb trees at various angles without causing damage.

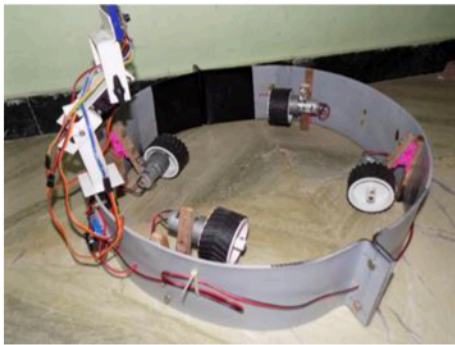


Fig11. Robot Frame Design

The motors are controlled by an L298N motor driver circuit, which allows for forward and reverse rotation. The ESP32 Cam module provides control signals via GPIO pins, while MG996R servos receive PWM signals for precise rotation.

The 3-DOF arm shown in figure 12. mounted on the robot has specific lengths for each segment, ensuring accurate movement and positioning. The arm's design, combined with inverse kinematics, allows it to recognize objects and perform tasks such as cutting palms.



Fig 12. 3-DOF Arm.

The hardware design of the multi-degree-of-freedom

robotic arm incorporates advanced components and precise calculations to ensure accurate, reliable, and efficient operation within the Gazebo simulation environment and in real-world applications.

5. Conclusion

This innovative robot leverages a minimalistic design approach, utilizing the least number of components to ensure efficiency and cost-effectiveness. Compared to previous models, this robot offers a substantial reduction in both cost and complexity, making it an accessible solution for a broader range of users. The inclusion of a camera allows the robot to monitor obstacle distances and navigate accordingly, ensuring safe and precise movements. The robot is equipped with four motors connected to a controller, providing the necessary RPM for vertical movement. This configuration enables the robot to climb trees efficiently, even in challenging weather conditions. The robot's design and functionality have been proven effective in significantly reducing the incidence of death and injuries associated with tree climbing. Its ease of control and robust performance in critical climate situations make it a reliable tool for maintaining safety. As a prototype, this model serves as a proof of concept, demonstrating its potential for practical application. The combination of cost-effectiveness and high functionality positions this robot as a valuable asset in addressing the hazards of tree climbing. In conclusion, the palm-climbing robot represents a significant advancement in robotics, offering a practical solution to a dangerous task. Its development underscores the potential of robotics to improve safety and efficiency in various fields, paving the way for future innovations and enhancements.

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None

Author contributions

All Authors equally Contributed

Conflicts of interest

The authors declare no conflicts of interest.

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